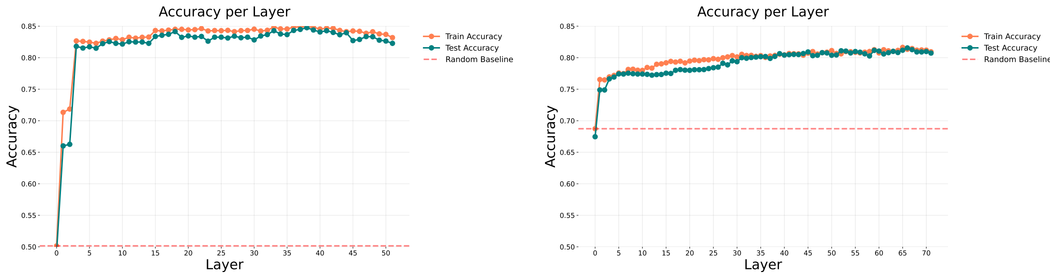

(a) Gemma 2 2B

(b) Pythia 12B

Figure 4: Layer-wise linear probe accuracy for Gemma 2 2B and Pythia 12B. Orange/blue: train/test; red dashed: random baseline.

4.2 Robustness Against Context Perturbation

To what extent is a model's factual self-awareness robust to changes in input context? To address this question, we train linear probes on each model layer and evaluate their performance on contextually modified input samples. We design four targeted experiments to systematically assess the test-time robustness of self-awareness directions under such perturbations.

Quotation Marks. Enclose the entity name within single or double quotation marks:

relation attribute

The director of the movie 'Inception' is Christopher Nolan.

entity type entity name

Prompt formatting, including punctuation, spacing, and quoting, has been shown to significantly impact model performance (Gonen et al., 2023; Sclar et al., 2024).

Statement Question. Prepend a natural language question crafted from the sample quadruples:

relation attribute

Who is the director of the movie Inception? The director of the movie Inception is Christopher Nolan.

entity type entity name

Rephrasing inputs as questions can improve model reasoning and accuracy, with even minor phrasing changes affecting outputs and models performance (Kojima et al., 2022; Mizrahi et al., 2024).

Few-Shot. Prepend few-shot context by adding a small set of sample (e.g., three) from the dataset to the input. These samples are chosen according to one of two entity modes.

Only: all selected samples have the same entity name; however, the relation and attribute may vary:

relation attribute

The release year of the movie Inception is 2010. The director of the movie Inception is Christopher Nolan.

entity type entity name

Unique: each entity name appears only once in the context; however, the relations between entities are not necessarily distinct.

relation attribute

The genre of the movie The Matrix is science fiction. The director of the movie Inception is Christopher Nolan.

entity type entity name

Few-shot prompting improves generalization but remains brittle to surface-level changes, highlighting the need to assess robustness (Sclar et al., 2024).

Random Statement. Prepend a fixed, unrelated grammatically correct sentence to the input prompt: "The cat darted under the couch as the thunder cracked outside."

relation attribute

The cat darted under the couch as the thunder cracked outside. The director of the movie Inception is Christopher Nolan.

entity type entity name

Semantically neutral distractors, such as irrelevant prefixes, significantly affect model predictions, indicating sensitivity to prompt framing beyond content (Sclar et al., 2024).

We assess the robustness of self-awareness directions to contextual perturbations at test time using the Gemma 2 2B model, shown in Table 2. The model is trained on a fixed dataset, and its performance is evaluated across various input modifications.

Adding quotation marks—either single or double—yields only minor reductions in test accuracy (0.802 and 0.797) compared to the unmodified baseline (0.820), indicating robustness to superficial punctuation changes. Rephrasing factual prompts as questions (*Statement Question*) leads to a larger performance drop (0.756), suggesting sensitivity to structural semantics beyond lexical form. In contrast, appending unrelated content (*Random Sentence*) minimally affects performance (0.791), indicating resilience to distractors when the core entity–relation structure is preserved. The *Few-Shot*

Table 2: Self-awareness directions robustness against various context perturbations for Gemma 2 2B model. Δ indicates test accuracy gain over the random baseline.). Standard deviation in parentheses.

Modification Type	**Train (shared across all)**			**Test (varies by modification)**		
	Loss	**AUC ROC**	**Accuracy**	**Loss**	**AUC ROC**	**Accuracy**
None				**0.397** (0.064)	**0.896** (0.060)	**0.820** (0.056)
Quotation Marks (single)				**0.431** (0.066)	**0.882** (0.057)	**0.802**(0.060)
Quotation Marks (double)				0.448 (0.066)	0.877 (0.056)	0.797 (0.072)
Statement Question	0.383 (0.063)	0.901 (0.061)	0.833 (0.052)	0.520 (0.092)	0.856 (0.056)	0.756 (0.061)
Few-Shot (Only)				0.708 (0.183)	0.771 (0.069)	0.650 (0.083)
Few-Shot (Unique)				0.538 (0.157)	0.845 (0.074)	0.753 (0.065)
Random Sentence				0.458 (0.070)	0.871 (0.057)	0.791 (0.061)

(Only) setting causes substantial degradation (0.650), likely due to signal dilution, whereas *Few-Shot (Unique)* better maintains accuracy (0.753), underscoring the role of relational diversity in preserving linear decodability.

Overall, these findings underscore that factual self-awareness in LMs is relatively robust to superficial noise but sensitive to semantically meaningful shifts in input structure.

4.3 Impact of Sampling Parameters k-l on Probe Behavior

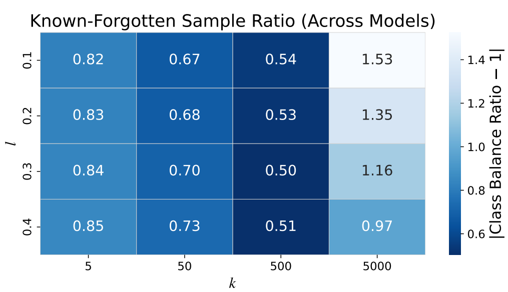

Figure 5: Known-Forgotten sample ratio for each (k, l) configuration, aggregated across all models. Lower values (darker) indicate more balanced retention, helping identify the globally optimal (k, l) setting that generalizes across models.

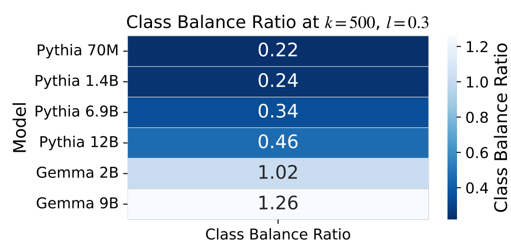

Figure 6: Class balance ratio at $k = 500, l = 0.3$ for each model. Values closer to 1.0 indicate more balanced retention, though Gemma 2B diverges significantly, suggesting this configuration may not generalize well to all architectures.

We systematically evaluate the effect of varying k and l values on linear probe performance and the class balance between `known` and `forgotten` samples for Gemma 2 (2B, 9B) (Team et al., 2024) and Pythia (70M, 1.4B, 6.9B, 12B) (Biderman et al., 2023) models. Known-to-forgotten class ratios for different k–l configurations per model m are listed in Appendix D, with visualizations in Figures 13 and 12 for Pythia and Gemma 2, respectively. Among the Pythia models, Pythia-12B exhibits the highest known-to-forgotten ratio under the default k–l setting, with $\text{Ratio}_{\text{Pythia-12B}}(k = 500,\ l = 0.3) = 0.46$. In the Gemma 2 series, Gemma-2B shows a substantially more balanced attribution of factual knowledge, with $\text{Ratio}_{\text{Gemma-2B}}(k = 500,\ l = 0.3) = 1.02$.

To complement the analysis of class balance, we examine how the (k, l) configuration affects downstream recovery of factual self-awareness in terms of linear probe performance. In Figure 11 in Appendix D, we report the test and train accuracy improvements over a random baseline for the two representative models—Gemma 2B and Pythia 12B—across all evaluated (k, l) settings. These heatmaps reveal how performance varies with sampling parameters, where each cell shows the absolute accuracy gain (test/train) above random guessing.

Notably, linear probes on the Gemma 2B model exhibit consistent and substantial gains across multiple configurations, particularly around $k = 500$, whereas Pythia 12B shows smaller but more stable improvements. These results highlight the trade-off between class balance and discriminative performance, further motivating the choice of $(k = 500,\ l = 0.3)$ as a configuration that yields competitive accuracy.

5 Scaling Behavior

To further investigate the emergence of self-awareness directions in language models, we analyze models from the Pythia scaling suite Biderman et al. (2023), focusing on variation across model sizes—specifically 70M, 1.4B, 6.9B, and 12B parameters.

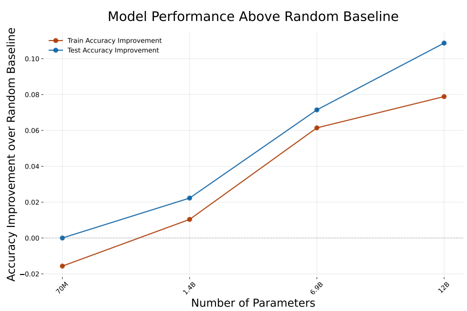

Figure 7: Accuracy gain over random baseline for training (orange) and test (blue) linear probes as a function of model size. Larger models exhibit greater gains in accuracy, with test performance benefiting more substantially from scaling.

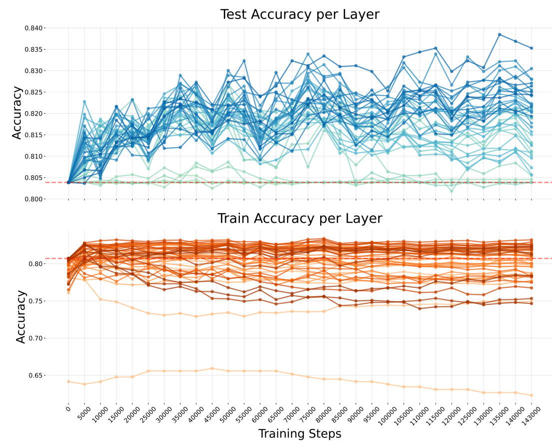

Figure 8: Linear probe accuracy across Pythia 1.4B training checkpoints/tokens. (Top) Training accuracy by layer (warmer colors = deeper layers). (Bottom) Test accuracy by layer (cooler colors = deeper layers, brighter = later layers). Red dashed line: random baseline.

We measure the emergence of linearly decodable features associated with self-awareness by computing the accuracy improvement of linear probes over a random baseline across models of increasing scale. As shown in Figure 7, both training and test accuracy improvements grow monotonically with model size, indicating that larger models develop more robust and generalizable representations. Notably, the improvement is more pronounced on the test set, suggesting that increased capacity enhances the transferability of these features beyond the training distribution.

To further examine how these features evolve during training, we evaluate checkpoints of Pythia 1.4B, spaced every 5000 steps from 0 to 143000, as shown in Figure 8. At initialization (`0`), linear probe accuracy is at the random baseline, indicating no self-awareness directions in the untrained model. During training, middle layers consistently yield the highest probe accuracy, while early and late layers perform worse. Training accuracy rises quickly and plateaus early, while test accuracy improves more gradually. On the test set, the highest accuracy occurs in the upper layers, suggesting self-awareness features are more strongly encoded later in the network. In contrast, middle layers dominate on the training set, indicating a divergence in the distribution of generalizable vs. task-specific features across depth.

6 Conclusion

In this work, we explore the landscape of factual self-awareness of pretrained LMs. We precisely ask the question whether an LM encodes its certainty within its neural representations that it will be able to recall a given factual association. We frame this as awareness-at-generation. Providing an affirmative answer, we show that encoding of such a signal is linear and surprisingly robust against context perturbation. We find that while a threshold model size is essential for the self-awareness signal to appear, the strength of the signal is not directly proportional to scaling. Same trend is observed in terms of training scale: the signal appears quite early in training and saturates quickly. We argue that this specific type of self-awareness, which is evident at the time of generation, can serve as a stronger entry point to curb LM hallucination compared to post-hoc truthfulness, investigated hitherto. Compared to the latter, this awareness-at-generation can be repurposed to restrain the model from a generation attempt before the actual generation.